Techné: Research in Philosophy and Technology ISSN: 1091-8264
20:3 (2016): 177–197 DOI: 10.5840/techne201671949

Beyond Technological Mediation: A Normative Practice Approach

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Abstract: Several philosophers of technology have argued that technology mediates human actions. For example, in the branch of post-phenomenology, authors such as Don Ihde and Peter-Paul Verbeek have described the mediating aspects of technology in terms of morality of technology (more prominent in Verbeek) as well as in the sense that technology changes our perception of ourselves and the world (more prominent in Ihde). In this article, different existing types of mediation are presented, critiqued, and enriched. The four types are illustrated by referring to military high-tech environments with a focus on visual data and imaging technologies. These technologies can mediate actions (1) by inviting certain behavior, (2) through amplification and reduction, (3) through built-in norms, and (4) through interpretation. The four types of mediation mainly focus on the technology or technological artifact itself. What these approaches fail to grasp, however, is the specific user practices in which most technologies function. In this article, it is argued that to understand the mediating aspects of technology more fully, attention should be paid to the specific user context in which the technology functions. Therefore, an enriched understanding of the four types of mediation of technology is proposed by taking the lens of normative practices and analyzing the different types of mediation through this lens. The Kunduz Airstrike Incident, which took place in 2009 in Afghanistan, is a case in which a visual data sharing device called Rover played a prominent role. This case is used in this article to illustrate how technology mediates human actions in military practice.

Key words: technological mediation, normative practices, Kunduz airstrike, visual data sharing, moral significance of technology, reformational philosophy

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1. Introduction

A colonel in Afghanistan, together with his sergeant, who was a Joint Tactical Air Controller (JTAC) trained to guide the pilots prior to and at the time of an airstrike, interpreted dots on a screen—provided by live video feed from a fighter jet flying over the banks of the Kunduz River—as insurgents. The colonel assigned the status of "imminent threat" to the events on the ground and ordered an airstrike. Sadly, after the bombs had dropped, the majority of casualties turned out to be civilians (Gebauer and Goetz 2010).

This case, known as the Kunduz Airstrike Incident, and its serious consequences forces us to re-think the normative aspects of military technology. It provides a context for philosophical analysis of the use of technology on the battlefield. Several philosophers of technology have offered a framework for understanding the moral significance of technology. For example, Peter-Paul Verbeek (2006) offers a framework of technological mediation in his attempt to do justice to the role of technology in ethics and morality. My aim is to go beyond this framework in order to grasp the moral impact of technology on the battlefield. This article, therefore, does not deal with military technology as such, but focuses on the role of technology—in particular, network enabling technology—on the battlefield. Analyzing this case from an enriched technological mediation perspective reveals important insights about the use of technology on the battlefield, especially the use of networking and information and communication technologies (ICT).

The Kunduz airstrike case is also an example of what is happening more often in modern military missions: soldiers are being equipped with information systems through which they are supposed to receive and send information over a network, which connects other soldiers and non-military actors. Technology is usually perceived as a neutral enabler or facilitator of information exchange in such network enabled operations (NEOs). Although technologically sophisticated (visual) aids are designed for reducing casualties, what this case makes clear is that introducing them into a high-stakes decision environment—such as airstrike scenarios—can lead to enormous and harmful consequences. In the case of the Kunduz airstrike, the interpretation of visual data by members of a military network, amongst other factors, resulted in the killing of many civilians. Often, it is the case that people who develop and use these technologies are unaware of the mediating aspects of devices such as Rover (the device that provided the grainy images in this case). It is just too easy to say that the commander who ordered the airstrike should have known better what was going on in reality at the Kunduz

riverbanks and never should have ordered the airstrike on the basis of grainy images on a screen. For the commander, the grainy images on the screen were an important part of his reality. Therefore, analysis of how individuals experience technology and the world around them as depicted by technology is much needed and several philosophers of technology have conceptualized this human-technology relationship. What scholars thus far have failed to address is that often technologies and their users are embedded in a specific social structure, which I will refer to as *normative practice* (more on this will follow). The normative evaluations of technology have so far focused either on what users (or designers) should or should not do with technology, resulting in ethical guidelines or phenomenological approaches about how the user experiences the technology, or on the technological artifact itself by adopting an analytic approach to, for example, norms in artifacts. Both approaches have genuinely revealed normative aspects of technology—some of which are addressed in this article—but these approaches, including the technological mediation thesis (more on this in section 2), alone cannot fully explain the moral impact of technology.

Some existing theories, mainly in the field of organizational science, call attention to the relationship between technology and its mediated user context. For example, structurational perspectives (Giddens 1984, Orlikowski 2008, DeSanctis and Poole 1994) or Actor Network Theory (Latour 2005) look at technologies in practice: how technology and the organizations in which technology functions mutually shape both technology and the structure in which technology functions. However, these theories, at large, abstain from pointing out the normative issues at stake and remain silent about the desirability of certain technological developments in, for example, healthcare organizations.

Some philosophers of technology have paid attention to how technology or technological artifacts help shape specific practices on a "macro-level" in science (Latour 1987), politics (Winner 1980, Winner 1988), or societies (Feenberg 1991). I argue that by revealing the normativity that is inherently present in the specific "meso-level" user practices (for example: healthcare, business, military practices), one can get a better grasp of the moral role of technology in that specific practice. In this article, concepts found in reformational thinking[1] (for example, the distinction between structure and direction) are used in order to fill this gap. Throughout this paper, the military practice will be used in addressing the normative dimension of networking technologies, in their context.

2. Mediation Theory: Four Types of Mediation

The claim that technology mediates human action and therefore has a moral dimension is not entirely new. Günther Anders (1987) once described how technologies affect our empathy when he discussed the bombing of Hiroshima. To drop a bomb, the pilot only has to push a button. He neither sees the face of the victims, nor the consequences of his action. Without seeing or hearing the impact, he is able to kill millions of people, while, as Anders claims, listening to classical music (Anders 1987). The same kind of argument is made for contemporary drone pilots, who operate with similar technologies to Rover from the Kunduz airstrike. They are in Nevada sitting behind a screen in a cubicle, observing potential insurgents. Upon request, they perform airstrikes thousands of miles away from their targets and go to the safety of their homes for dinner after eight hours of work. The way in which the moral dimension of the battlefield is being presented, perceived, and "digested" is altered by these technologies.

Before carrying on with the analysis of the way people experience themselves and the world around them in a highly technological environment, I assert that I will not deal with specific behavior or the way people reason about moral issues here. That is the task of the psychologist and moral philosopher, who respectively search for general psychological processes that underlie the behavior that can be registered in experiments, and try to understand the nature of deliberations concerning right and wrong. The aim in this article is of a different nature, namely to bring to light the context in which technology functions when evaluating the moral dimension of technology. In modern military missions, people often have to make high-stakes decisions. These decisions will be influenced by the way the technology has shaped their interactions with others and their environment.

Verbeek argues that neither humans nor objects exist in isolation, but that they always exist in relation to each other. He refers to this relation as technological mediation: technological artifacts can direct our actions and experiences and therefore play a role in moral deliberations. According to Verbeek (2009), the "ensembles" of humans and technology possess moral agency. He claims that moral agency should be seen as a result of the interaction between humans and technology rather than as an intrinsic property of humans as in traditional philosophy. Technological mediation theories fail to grasp that technologies are often embedded in predefined social structures, which I refer to as normative practices. I argue that the moral dimension of technology should be understood in light of the normative practices in which technologies often function. The ensembles of

humans and technology are not enough to understand the way in which technological artifacts steer behaviors. Human beings are moral agents who often decide and act within normative structures. It is not merely technology that steers and co-constitutes moral actions, but both normative structure and technology (and psychology, as I have argued in Boshuijzen-van Burken and van Bezooijen 2015) are of moral significance for human actions.

In the following subsections I discuss four different ways in which technology, in particular visual data sharing devices of which Rover is an example, mediates human actions. Technology mediates actions through inviting certain behavior, through amplification and reduction, through built-in norms and through necessary interpretation. Technological mediation is contextualized and its moral relevance is evaluated in light of the normative practice in which Rover functions.

2.1 Technology Invites Action

The first type of mediation of technology in this section is that technological artifacts invite certain behaviors. Pols (2011) distinguishes between artifacts inviting (cf. Verbeek 2008a) and prescribing (cf. Akrich 1992) actions.[2] Following Verbeek's (2008a, 2008b) approach to technology, technology not only poses new moral problems (for example to request an airstrike through which people will be killed, or to refrain from doing so), but the way technology represents an actual situation influences the way we evaluate the problem. Thus, technology invites us to make a choice in a certain direction. Technology mediates moral decisions, because, in the case of Rover, it not only shows the person that is caught by the camera's lens on a screen, but it depicts the person on the Rover screen in a specific manner, thus influencing one's evaluations and consequently one's decisions concerning the person on the screen. This is also the case when visual imaging technologies reduce friend and foe to merely objects, distinguishable as moving black or gray dots on a screen, as Rover did during the Kunduz Airstrike. This approach to the mediating aspects of technology, however, neglects that technology mediates in a very specific practice. Understanding the way in which technology mediates moral decisions in these structures is necessary for evaluating the normative aspects of technology. Decision-making structures that to a great extent define the (correctness of) actions are already in place, prior to introducing of technology into the practice. It does not mean that these defining structures have become redundant and play a lesser role, as if solely technology are steering actions. The mediation approach is right in that technology can create new moral problems and invite behaviors, such as pulling a trigger. However, moral problems are often

practice-specific: cutting someone's belly open is more often morally problematic for a bus driver, than for a surgeon. The context in which technology is used decides what set of actions is available for the user.

2.2 Reduction and Amplification in Technology

Rover is a technological device for sharing visual data. The technological mediation of perception is said to work through the mechanisms of "amplification" and "reduction." In other words, technology enhances what is often difficult or impossible to see with the bare eye, yet also weakens or hides what cannot be captured by the mediation technology. Don Ihde reminds us that: with every amplification, there is a simultaneous and necessary reduction. And the amplification tends to stand out, to be dramatic, while the reduction tends to be overlooked (Ihde 1979, 21).

Rover has an option for zooming in and out and one of its features is a thermal camera for night vision. In the thermal camera mode, it only shows what is within a certain temperature range and therefore amplifies things that would not even be visible with the naked eye (we cannot "see" temperature in a direct sense). With a Rover device, one can see more, especially at night, however, there is a reduction at work as well. The reduction is that the objects that are presented on the screen are reduced to merely images corresponding with temperature variances (leaving out color, fabric, expressions, size, etc.). What is being presented or not depends on its (or his or her) temperature. Through this mechanism of amplification and reduction, it imposes and/or removes (moral) significance: a truck engine may look more significant than a child, since its "hot-surface" is more profound than that of a child. Therefore, the presence of a truck is amplified on the screen, while one needs to look closely to notice the presence of a child on the screen. Also, what happens outside the scope of the thermal camera is hidden from the eye of the observer and therefore, often and understandably neglected. This insight is certainly helpful in pointing out the mediation aspects of technology through amplifications and reductions; however, it does not say when and how this amplification or reduction is morally significant. For a normative evaluation of the role of this mediation technology, one needs reference to a specific user context. This type of mediation (reduction and amplification) in Rover may be morally problematic for a commander, but it may not be problematic for a pilot. A commander may be distracted by the amplifications; whereas, for a pilot, the amplifications are helpful in distinguishing various objects prior to an airstrike.

2.3 Technology Has Built-in Norms

The third type of mediation is related to built-in norms in technology. A majority of imaging technologies operates on the basis of underlying software algorithms. Felicitas Kraemer, Kees van Overveld, and Martin Peterson (2011) have argued that some algorithms are value-laden. For example, designers of detection technologies often make value-judgments when programming an algorithm resulting in differences in, for example, detection thresholds, offsets, color, exemplifications, sensitivity, etc. that are seemingly external factors. Kraemer, van Overveld, and Peterson (2011) use the example of tagging cancerous cells in medical image analysis. Depending on the software designer's preference between avoiding false positives and false negatives, a certain threshold is chosen in the algorithm for detecting diseased cells. The threshold level is a built-in norm to decide whether a cell counts as good (healthy) or bad (diseased). What Kraemer, van Overveld, and Peterson (2011) recognize is that preferring false positives over false negatives (or vice versa) depends on the operational setting in which the detection tool is used. This sensitivity to the context is very helpful in understanding the technological mediation. For Rover, the soldiers have the option to choose between a "white hot" or a "black hot" algorithm in the device and they usually determine on the fly whether they choose to present everything that is "hot" in black and everything "cold" in white or vice versa. However, depending on the context, this option could lead to a hidden bias; think, for example, of the possibility of using the tool around sunset, when the earth has heated up and some materials (roofs, rocks, cars) radiate heat longer than others, which may change the contrast of equally important items. It may even lead to an increase of false positives or false negatives, depending on the chosen setting. Taking into account the context in which Rover is used (for example when Rover is used in an office or outside in bright sun) helps determine which algorithms may cause problems or unwanted biases in the case of Rover and how this may cause problems or not. The norms for what counts as "hot" (for example, a human being) and what counts as "cold" (for example, a road) depend on the chosen thresholds.

2.4. Mediation Through Translation and Interpretation

For the fourth type of mediation of technology, the work of Ihde is employed. He speaks of a hermeneutic relation of technology with its users or the world (Ihde 2002). Hermeneutic relations are relations in which the technology gives a representation of the world. An example is the thermometer. A traditional thermometer consists of a liquid column in a glass tube with a calibrated scale marked on the

tube. In order to know what the temperature is, one looks at where the column ends at the scale. The technology mediates through translating temperature to a scale or a number on a digital device. The hermeneutic dimension of mediating technology refers to a necessary interpretation of what is presented as a liquid column or depicted by a number in case of a digital thermometer, before understanding the representation on the screen or scale. The numbers on the scale stand for degrees Celsius or Fahrenheit and need to be interpreted as temperature. For some of the applications of imaging technology, a highly specialized or critical hermeneutical "reading" is needed to understand the meaning of what is presented (for example, in spectroscopy). For the case of Rover, it may seem obvious how to read the screen and how to interpret the thermal dots, but in reality, this is not, as the Kunduz Airstrike Incident made clear. The first level of interpretation is the distinctions between the dots, which could denote cars, people, or weaponry for example. The second level of interpretation concerns the meaning of the dots in a normatively different sense. The normative interpretation of the dots went dramatically wrong, because of the (mis)interpretation of the dots as insurgents instead of civilians.

Ihde's insights into the hermeneutic dimension of imaging technologies is helpful, because they offer a way of understanding technological mediation, but they have not so much to offer in terms of a normative evaluation of this type of mediation. Ihde seems to overlook that evaluating the correctness of the interpretation is related to the practice in which the mediating device is embedded. A practice has certain norms and values to it and even embodies a certain worldview (more on this in section 4), which may affect what counts as a correct interpretation of what is depicted. Problems may even arise if no univocal hermeneutical reading exists within the practice or amongst different practices that collaborate around the same visual technology.

3. Problems for Technological Mediation Approach

Verbeek seeks to answer the question of the moral dimension of technology in the human-technology relationship itself, and many criticisms have been brought up against that point. For example, Martin Peterson and Andreas Spahn (2011) or Peter Kroes (2012) critique Verbeek's understanding of the moral agency of technical artifacts and they attempt to (re)focus on the agency of the user who uses the technology (without denying that technology is of moral significance). The proposal in this article is of a different kind, but has some similarities with Christian Illies and Anthonie Meijers's (2009) proposal to evaluate the moral significance of

technology in view of how artifacts affect the *set of actions* available to the agent (they call this *Action Schemes*). It is a shift in perspective, away from looking at how artifacts influence individual actions, toward a wider focus on agents and the context in which they use technology. According to them, actions are always embedded in a dynamic *context* and the Action Scheme is relative to this context (Illies and Meijers 2009, 427). In this article, I give a further theming to this context. The evaluation of the actions done with technological objects is determined by the normative practice in which the action takes place. It is a thematization of the rules and regulations that are in place in the context in which people use technology. These rules and regulations promote certain actions and discourage others.

The traditional, moral question of the man-with-gun and man-without-gun should be understood with reference to the context in which the man-with-gun is embedded. The man-gun association by itself is insufficient to evaluate the moral dimension of the gun. Verbeek is right when he considers the gun as mediating moral actions (killing another person), but the question of morality should not be solved with reference to the man-gun relationship alone, but with reference to the social structure in which man-gun ensemble stands. The normative practice determines the morality of killing another person. Shooting another person is not necessarily morally wrong and can in some contexts be the morally right thing to do, depending on the normative practice in which the action takes place.

Moral agency should be understood in the context of the normative practice in which the technology and user are embedded. Of course, there are also technologies that function outside normative practices, but very often, they are embedded in a context of professional actions. Conceptualizing the context for moral decision-making in terms of normative practices does not avoid the problem of moral agency, but does justice to the reality in which technology and humans often function. In the following section, a recapitulation of the normative practice concept is given, which is the key concept that serves as a means to theming the context for technological mediation.

4. Normative Practice Approach

In this section, technological mediation and its different forms is taken a step further, namely by focusing on embedding of technology in social practice, rather than focusing on the individual or technology itself to understand the moral aspects of technology.

The perspective of "normative practices" is a conceptual tool that clarifies the specific context in which technology is often used. Often, people who work with

technology have entered a social and organizational structure that already exists and in which certain rules, norms, procedures, etc. are in place. People are not as free as one may wish in using technology (in the sense that people can make choices as they like with regard to the use of technology), but that there are rules and norms in place on how to use technology. This may influence the way technology is deployed and therefore, matters for an evaluation of the moral dimension of technology. Verbeek's mediation thesis reveals the profoundness with which technology affects moral actions, but it is not convincing in the way it conceptualizes the moral role of technology.

Henk Jochemsen and Jan Hoogland (1997) were inspired by Herman Dooyeweerd and have argued that entities have a qualifying aspect, which determines the intrinsic normativity of the entity (see also Strijbos 2006a, Strijbos 2006b, Jochemsen 2006, Basden 2008, Basden 2011). A qualifying aspect is what defines the existence of an entity and gives it its purpose. Brigitta Bergvall-Kåreborn, in the same line of inquiry, states that the qualifying function is a "concept that refers to the function that characterizes an entity, determines its inner structure, and makes us experience a specific identity in relation to this entity despite all the changes that may occur to it over the years" (Bergvall-Kåreborn 2002, 313). The approach in this article is in support of the views presented by Jochemsen and Hoogland (1997), but it also suggests an addition to their view: that professional actions that are performed in these normative practices cannot strictly be separated from technologies; often, technologies enable professional actions. For example, the chair of a judge is interwoven with the structural side of the legal practice, or in other words, there are rules about the position, occupancy, etc. of the chair. One cannot just sit in the chair of the judge, because sitting in that chair brings certain responsibilities and duties. The qualifying aspect of the chair of the judge is found in the juridical aspect: through the sitting of a person in that specific chair, certain juridical norms are being expressed and ought to be respected. The judicial practice even revolves for a great deal, at least in many cultures and throughout history, around the judges' chair. The person occupying that chair during a sitting of a court is the one whose verdict counts as binding. The verdict of the person in another chair in the room does not count as binding. Therefore, the chair of the judge plays an active role in moral actions (what one ought to do when sitting in that chair and what one not ought to do), but the core of the moral agency does not lie in the human-technology ensemble, but in the person who sits in the chair. The normative structure of the legal practice guides the evaluation of the moral actions. For another example, think of the patient's chair in the dentist's office,

which comes with its own set of norms. These are norms that hold for the aspect of (medical) care: the person who is sitting in that chair receives medical attention. The design of a chair often gives expression to the normative context in which the chair is used. The design of technological artifacts is, thus, influenced by the normative practice in which the technology functions. Technological design may support or discourage abiding by the rules and norms that tell what one should and should not do with the technology.

In the case of the military practice, this may be more apparent. Technology can be employed in a way that it (dis)respects the norms that hold for the specific practice. This is the case if, for example, the Rover device is used for committing genocide or controlling rather than guiding a pilot in executing an airstrike. The structural rules are about the relevant aspects that hold for the practice and they can be applied in a good or in a bad manner. Technology is often interwoven with the rules and norms; in the following section, some examples are given of how this is the case for Rover. Different practices and even sub-practices exist (for example, the pilot practice and a reconnaissance practice: they have differing rules and norms and procedures they work with, as well as differing objectives and goals). This means that one and the same technology can disclose as well as close down and hinder the flourishing of a specific practice, depending on the qualifying function of that (sub)practice. I will now show that the normative practice view takes into account the differences in *structure* and *direction* and allows for a normative evaluation of technology on the level of the user practice. The structural side of a practice relates to the norms (which can be tacit), rules, and procedures that enable and restrict actions in a practice. The directional side of a practice relates to worldview and ethos, or (unarticulated) motives that drive a practice. The different types of mediation of technology that were discussed in the first part of this article focused on the artifact itself and did not account for the user practices, with different structures and directions, in which the technology was used. In the following section, some additional insights that follow from a normative practice approach are addressed by making use of the Rover example with respect to structure and direction.

5. Beyond Technological Mediation in the Case of Rover

The moral significance of technology can be shown in a two-fold manner, namely with a focus on the technological artifact itself and with a focus on the context in which the technology is used, a specific practice. For both explanations, one can employ the distinction between structure and direction. The technological artifact

itself has a structural side: there are limitations and characteristics that are constitutive for Rover—for example, the guiding principles are to support the JTAC in determining a target in a univocal manner. Other principles that hold are, for example, thermo-radiation principles for Rover, or aesthetic principles which one can ignore or respect, resulting in a better or worse aesthetic design. Technology also has a directional side, which refers to the underlying worldview: for Rover, this could be the view that collateral damage (harming local people or infrastructure) during airstrikes is bad and must be reduced by gaining optimal "situational awareness" using thermal cameras that are linked to different parties in the network. The directional side guides the realization of the structural side—for example, designing or rolling out this system throughout the organization to optimize situational awareness on all hierarchical levels (as opposed to providing Rover only to a very specific user group).

The second line of thought relates to the qualifying function of the technological artifact in the perspective of the user practice. Technologies can disclose (cf. Strijbos 2003) and thereby create room for the realization of the specific (moral) value inherent in a practice. The Rover device in the military practice can leave room for realizing the promotion of justice, by using it to make better distinctions prior to the use of force. However, technology might form an obstacle for the normativity of the practice to be realized, and in the case of the military practice, it might hinder the promotion of justice[3] through the employment of thermal cameras in specific sub-practices. This may be the case when Rover is employed to control people rather than support soldiers in their daily work. The deployment of Rover may also cause communication chaos, since more people want to have a say on what they see on the screen. Streamlined communication is vital for the qualifying function of the military practice: often, decisions about life and death need to be made in demanding circumstances and, therefore, discussions with multiple partners about what is happening on the screen can have disastrous consequences as we have seen in the Kunduz case. The normative practice view can add valuable insights when interwoven with the four types of mediation that were presented earlier. The normative practice view is not an addition in the sense that it is a new type of mediation of technology, next to the types that were presented earlier, but the four different types of mediation of technology are enriched by viewing them through the lens of normative practices. It reveals that normative issues related to working with these technologies are often dependent on the user context.

5.1. Rover Poses Different Questions and Invites Different Behaviors

Analyzing the moral dimension of Rover in light of Verbeek's mediation thesis gives us the insight that the use of Rover raises new (moral) questions and problems and invites certain behavior. The normative practice model, however, reveals that what counts as a "problem" may differ and depend on the qualifying function of the practice in which Rover is used. Also the moral evaluation of the problem may differ, depending on the (sub)practice in which Rover is used. For example, a Rover device gives the possibility to view trucks that are hijacked on a riverbank, such as was the case in Kunduz. A view on the trucks would not have been possible without Rover, since it was 2:00 o'clock in the morning. Therefore, due to the presence of Rover, a moral issue arose for the commander, namely to bomb the trucks or to leave the scene, running the risk that the fuel trucks would be used in a suicide attack against his camp. Rover, showing black dots next to trucks, also may have invited the bombing, because it would mean that a host of insurgents could be eliminated at once. These inviting aspects of Rover depend on the normative practice in which the technology functions. For example, Rover will not be inviting for the pilots, because it is not in the structure of their normative practice to bomb a truck upon their own initiative. Another example is the moral decision to either bomb the fuel truck directly, or to bomb the riverbank next to the trucks. This issue can be evaluated differently, depending on whether one reasons from the perspective of a commander or from the perspective of a pilot. The practices of the commander and the pilot have different structures and directions. Rover may be very helpful and contribute to the qualifying function of the pilot practice—for example, to gather insights with regard to avoiding collateral damage. On the other hand, it may negate the commander practice: focus on the details may distract from keeping general oversight, which is typically what the commander is there for and which is the guiding principle of the commander. The Rover device provoked the commander in Kunduz to behave badly (killing civilians); while for the pilots, it invited moral behavior (a low flight over the people to scare them before the actual strike). This moral evaluation of Rover cannot be spelled out by Verbeek's mediation thesis alone, but one needs the normative structure of the practice in which the technology is used.

In the Kunduz Airstrike Incident, Rover connected different practices, which led to disagreement between the commander and the pilots over the course of action followed, after the commander assigned the dots on the screen as an "imminent threat." This was due to the difference in norms and procedures for the

different practices. A commander's practice focuses on keeping clear oversight over the operation. He or she probes when it is (not) allowed to engage according to Geneva Conventions and when it is appropriate to request air support in the first place. Within the structure of the commander, there is no guideline for handling data received through this Rover device. In other words, the structure of the commander practice did not account for this novel technology. The pilot, together with a JTAC, decides about the details of the airstrike, such as where to bomb exactly and with the least collateral damage. The Rover device can be of great help with this task. A pilot has to make a "gut-feeling check" as part of his procedure, meaning that in case the pilot doubts if the airstrike is according to the Rules of Engagement, he or she should withhold fire and abort the mission. The JTAC has an important function and is also called the "release authority." The JTAC and pilot, although lower in military rank, can refuse an airstrike that is ordered by a higher ranked commander.

In short, a commander requests an airstrike if he or she thinks it is appropriate. Rover was not designed to support the qualifying function of the commander practice. Therefore, it has invited inappropriate behavior in case of the Kunduz airstrike. Rover was designed for the pilot and JTAC practice, where it invited a different, more appropriate behavior.

5.2. Helpfulness of "Amplification and Reduction" Depends on Practice

The mediation principles of amplification and reduction are inherently present in visual aids such as Rover. The "Predator view" is a phrase used by tactical air controllers and describes what typically happens at the level of staff members at a headquarters (Call 2007, 72). They see the live video feeds on their screens and are sucked into the events that develop on the screens in front of them, forgetting about the bigger picture. The dangers of creating this tunnel-vision on the higher military levels are more poignant than on the lower levels, because the higher up in the military hierarchy, the more important it is to have an overview of the overall situation. The insights provided by technological mediation thesis show the problems that can arise due to amplifications and reductions, but mediation alone is not enough to determine the moral significance of amplifications and reductions. What the normative practice approach adds is that in some (sub)practices, the "addictive" aspects of moving images or a focused view of the Rover device may cause moral problems, while in other (sub)practices these amplifications and reductions are very helpful. Therefore, whether a certain technological device is morally desirable, depends on the practice itself. In some situations, the principles

of amplification and reduction are very helpful to make good decisions; whereas, in a different situation, they may hinder as well. Within the structure of the practice, there may be norms or procedures that create awareness for this reduction, but this may not always be the case. For example, part of the procedures in performing an airstrike, such as in Kunduz, is that a JTAC needs to have "eyes on" the target, meaning that the JTAC needs to be at a visual distance from the target (be it with binoculars). In modern military missions with more and more visual data and live feeds available, the "eyes on" criteria for performing an airstrike is met if the JTAC has a camera view on the situation. In Kunduz, the JTAC was not watching the riverbanks with binoculars, but through the live video feeds on the screen. The camera, which only gives a reduced view of reality, has recently entered the normative structure of the JTAC party. The "eyes on" criteria is one of the constitutive rules of the practice; guaranteeing safety for own troops, civilians, and local infrastructure is the guiding rule for this practice. This guiding principle may be hindered by mixing it up with the "eyes on" principle in case of (thermal) cameras, thus harming what the practice of JTACs is ultimately about.

5.3. Different Threshold and Value Settings Needed

Kraemer, van Overveld, and Peterson (2011) pointed out that imaging technologies mediate decisions through the value-ladenness of algorithms. The moral evaluation of this mediation, however, cannot be accounted for by reference to the value-ladenness of algorithms alone. The way in which thresholds matter in a moral sense needs a reference to the context in which the technology is used. The normative practice approach conceptualizes this context. In terms of structure and direction of normative practices, this means that the direction of the practice (for example, that no collateral damage is tolerated) guides the level to which the threshold of a thermal camera is set. If one wants to be absolutely sure that no living entity is present in a certain area (for example, prior to a bombing), the threshold for temperature in Rover may be set very low, in order not to miss out on anything alive. In the Kunduz Airstrike, for example, the threshold was set in such a way that both hot car engines as well as people were presented on the screen that presented the thermal images. But if the thermal camera is used within a different user practice—for example, the traffic controller practice—the threshold of the camera may be set very high, in order to be able to follow a certain convoy. The thermal camera then presents only hot car engines. These settings may differ for different situations, for example on the "technical level" (military jargon for the work-floor level), the accuracy of the device is important. Therefore, the thresh-

old levels may be part of the procedures that hold for the different practices. The Rover was designed for the purpose of supporting a soldier on the ground in his or her communication with the pilot about the exact location for the impact of a bomb. The Rover was used in this way and it supported communication between the JTAC and the pilots in the Kunduz case. However, very often the device is used by higher commanders in order to get a sense of control when tactical or even strategic decisions have to be made. A higher commander needs to get a general impression of where an incident unfolds and what more or less the scale of the incident is. A Rover device may be used for this, however, there is the pitfall of tunnel-vision, leading to harmful decisions, such as the killing of innocent people. From a moral perspective, the mediating role of the Rover device is problematic in the higher military level practices, because it may hinder good decision making.

5.4. Hermeneutic Rules in Structure of Practice

The hermeneutic mediation of technology, in which technology provides new representations in need of interpretation, itself is not sufficient to understand the moral role of technology. The standards for correct or incorrect interpretations arise within a specific practice of use and the moral effects of an interpretation can be large or will not matter much, depending on the context of use. A critical reading of a thermometer is often more morally laden in the case of a person who is seriously sick than in the case of deciding if it is a good temperature to go to the beach. A practice-oriented view on Rover conceptualizes the context of use of the device and recognizes that in the process of interpretation, a soldier already stands in a relationship with the Rover; Rover is embedded in the soldier's working practice, which has certain norms and values to it as well as a worldview. For example, the commander in the Kunduz airstrike was looking for ways to create a safe working environment for his subordinates, which influenced his interpretation of the events on the screen. He should have been aware of his eagerness to keep his own people safe, when interfering with the pilot's decision. The interpretation of the grey images on the Rover screen was driven by the commander's conviction to bring safety to his own troops and the region under his responsibility. This task cannot be cut free from an underlying worldview towards what safety is, what a good commander is, and how to bring stability in a region. Also, it should be noted that the commander in the case of the Kunduz airstrike had not been trained in interpreting the thermal images that the Rover presented, so he lacked sophisticated hermeneutical skills to interpret the images on the screen. The JTAC who was sitting next to the commander was skilled, or at least he was experienced in working

with Rover. Even though the hermeneutical interpretation may have seemed easy to do for the JTAC in the case of Rover, things still can go dramatically wrong due to a misinterpretation of what presents itself on the screen. Therefore, a view on mediation that takes into account the differences in specific hermeneutical rules that hold for specific user practices is needed. The normative practice view shows that differences in hermeneutical rules may cause potential problems when more people work together in a joint military operation.

6. Conclusion

In order to understand and evaluate the moral significance of technology, the notion of technology mediation has been applied and critiqued to a case with dramatic consequences and where technology clearly played a role. Verbeek and Ihde claim that technology mediates our perceptions and actions and thus can have profound effects on human beings, such as in the case of the Rover device that was used in the Kunduz airstrike. The main critique of the technological mediation thesis, however, is that it is not sensitive to the different shapes and forms in which technology mediates human actions and neither has it thematized the context in which technological mediation (often) occurs. A further categorization of the different ways in which Rover mediates human actions and perceptions was therefore made in this article.

The types of mediation were derived from post-phenomenology and other streams in philosophy of technology. These four types were chosen, since they contribute to a better understanding of military high tech environments with a focus on visual data and imaging technologies. It was shown:

(1) how technology can invite certain behavior (for example, an airstrike),

(2) how technology often mediates through amplification and reduction (for example, Rover imaging amplifies temperature, while important other aspects are reduced, such as color),

(3) that technology has built-in norms (for example, threshold levels for temperature determine what is visible on the screen or not), and

(4) that technology requires interpretation (for example, the dots on the screen of the Rover required a hermeneutic skill).

With these four different types, headed under the umbrella of technological mediation in this article, an attempt was made to create a moral understanding of technologies that enables the sharing of visual data on the battlefield. However, the

focus of these aforementioned approaches has been mainly on the technology or technological artifact itself; whereas, I have argued that it is more helpful to recognize its social context to be able to understand the moral role of technology on the battlefield. The practices in which technologies often function are normative and give guidance to the use of technological artifacts. Technology may alter or mix up the guiding principles of the practice, which sometimes leads to unfortunate decisions. For example, if the guiding principles for pilots are mixed up with the guiding principles for higher commanders, the information that is presented (through, for example, Rover) in the practice may be interpreted according to the rules and norms of another practice. Technology can be used in accordance with the guiding principles of the practice or not, which results in a flourishing of the practice or a hindrance of what the practice entails.

Evaluating the moral significance of technology from a normative practice perspective is not about judgments in the sense that technology is inherently good or inherently bad. Rather, it deals with the specific implications for the people who act and decide in the normative structure in which technology is embedded. Many accounts of the mediation of technology only focus on the artifact or the user. In this article, I have called attention to include a broader and deeper kind of moral evaluation of technological artifacts, which includes worldview, ethos, and an orientation towards certain values in addressing technology and its user context. It revealed how the use of visualization technologies affects decision behavior in military practices for better or for worse.

Notes

1. Reformational philosophy is a branch in Christian philosophy, pioneered by Herman Dooyeweerd and Dirk Vollenhoven in the 1930s.

2. I received further helpful comments from Auke Pols in this respect. He pointed at the distinction between real and perceived affordances, which he explains as follows: real affordances are the available options for acting with the artifact; perceived affordances are those actions that the user thinks she can do with the artifact. A good affordance is immediately visible as such; yet the "unusable" microwave also has (real) affordances, although you would not recognize them as such due to the bad design (they are not also the perceived affordances).

3. Elsewhere I have argued that the intrinsic normativity (related to the qualifying function) of the military practice is the promotion of justice (van Burken and de Vries 2012).

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